1) Define Ohm's law and write its limitations?

A) Ohm's law states that the current flowing through a conductor is directly proportional to the voltage across its ends, provided the temperature and other physical conditions remain constant.

$V = IR$

- Voltage = V (volts)
- I = current (Amperes)
- R = Resistance (ohms)

Limitations :-

1) Not valid for non-ohmic materials like diodes, transistors, electrolytes, etc.., because they do not obey a linear V relationship.

2) Temperature must remain constant if temperature changes, resistance also changes, and the law does not hold.

3) Not applicable to non-linear devices where resistance varies with voltage or current.

4) Fails in very high temperature or high electric field conditions, where the properties of the conductor change.

5) Not suitable for semiconductors, vacuum tubes, and filament lamps (their resistance changes when heated).

2) Explain kirchoffs laws with examples?

A) Kirchhoff's laws are 2 important rules used to analyze electrical circuits:

1) Kirchhoff's Current law (KCL)
2) Kirchhoff's Voltage law (KVL)

* KCL:-

The total current entering the function in a circuit is equal to the total current leaving the function.

It is based on the law of conservation of charge.

Eg:- At a function, three wires meet.

- current entering : $I_1 = 5A$, $I_2 = 3A$
- current leaving : $I_3$

Act to KCL,

$I_1 + I_2 = I_3$

$5A + 3A = 8A$.

* KVL:-

The algebraic sum of all voltages around any closed loop in a circuit is zero.

It is based on the law of conservation of energy.

Eg:-

consider a simple loop:

- Battery: 12V
- Two resistors & voltage drops are 7V & 5V

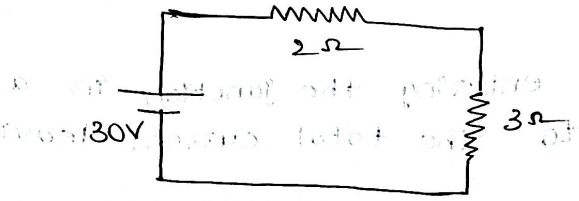
Act to KVL,

$(+12V) - 7V - 5V = 12V - 12V = 0,$

3) Problems on current division and voltage division?

A) Voltage division Rule:-

Find voltage across 2Ω & 3Ω resistors for a circuit shown in the figure?




Consider, $V_S = 30V$, $R_1 = 2\Omega$, $R_2 = 3\Omega$

$$V_{R_1} = \left[\frac{R_1}{R_1+R_2}\right]V_S = \left[\frac{2}{2+3}\right]30 = \left[\frac{2}{5}\right]\overset{6}{\cancel{30}} = 12V$$

$$V_{R_2} = \left[\frac{R_2}{R_1+R_2}\right]V_S = \left[\frac{3}{2+3}\right]30 = \left[\frac{3}{5}\right]\overset{6}{30} = 18V$$

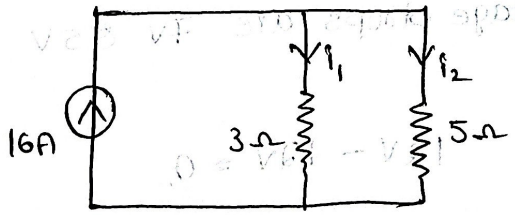
$$V_{R_1} + V_{R_2} = 12 + 18 = 30V_S$$

$$V_{R_1} + V_{R_2} = V_S$$

$$\text{i.e.} \quad -V_S + V_{R_1} + V_{R_2} = 0$$

Current division Rule:-

Find current passing through 3Ω & 5Ω resistors using current division rule:-




Consider, $i = 16A$, $R_1 = 3\Omega$, $R_2 = 5\Omega$

$$i_1 = \left[\frac{R_2}{R_1+R_2}\right]i = \left[\frac{5}{5+3}\right]16 = \left[\frac{5}{\cancel{8}}\right]\overset{2}{\cancel{16}} = 10A$$

$$i_2 = \left[\frac{R_1}{R_1+R_2}\right] i = \left[\frac{3}{5+3}\right] 16 = \left[\frac{3}{8}\right] 16 = 6A$$

$$i = i_1 + i_2$$

$$i = 10 + 6 = 16$$

$$i = i_1 + i_2$$

$$\therefore -1 + i_1 + i_2 = 0$$

4) problems on series-parallel combinations of RLC?

?.

5) Derive the eqn for Rms value, average value, peak factor and form factor for sinusodial waveform?.

A) * RMS value :-

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T [v(t)]^2 dt}$$

$$= \sqrt{\frac{1}{T}\int_0^T V_m^2 \sin^2(\omega t)\, dt}$$

$$= \frac{1}{T}\int_0^T \sin^2(\omega t)\, dt$$

$$= \frac{1}{T}\int_0^T \frac{1-\cos(2\omega t)}{2} dt$$

$$= \frac{1}{2}\left(1 - \frac{1}{T}\int_0^T \cos(2\omega t)\, dt\right)$$

$$= \frac{1}{T}\int_0^T \sin^2(\omega t)\, dt = \frac{1}{2}.$$

$$V_{rms} = \sqrt{V_m^2 \cdot \frac{1}{2}} = \frac{V_m}{\sqrt{2}}$$

$$\therefore \boxed{V_{rms} = \frac{V_m}{\sqrt{2}} \approx 0.7071\, V_m.}$$

* Average Value :-

The average of the waveform over one period is

$$V_{avg} = \frac{1}{T}\int_0^T V_m \sin(\omega t)\, dt$$

$$V_{avg} = \frac{1}{T}\left[-\frac{V_m}{\omega}\cos(\omega t)\right]_0^T$$

$$= \frac{1}{T}\left(\left(-\frac{V_m}{\omega}\cos\omega t\right) + \left(\frac{V_m}{\omega}\cos 0\right)\right)$$

$$\therefore \boxed{V_{avg} = 0}$$

* Peak factor :- (Crest factor)

Peak factor is ratio of peak to RMS :

$$Pf = \frac{V_m}{V_{rms}} = \frac{V_m}{\frac{V_m}{\sqrt{2}}}$$

$$= \sqrt{2} \approx 1.414$$

So

$$\therefore \boxed{\text{Peak factor} = \sqrt{2}}$$

* Form factor :-

form factor is ratio of RMS to mean of rectified waveforms.

$$\text{form factor} = \frac{V_{rms}}{V_{mr}} = \frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}} = \frac{V_m}{\sqrt{2}} \times \frac{\pi}{2V_m} = \frac{\pi}{2\sqrt{2}}$$

$$\text{form factor} = \frac{\pi}{2\sqrt{2}} \approx 1.1107$$

So

$$\therefore \boxed{\text{form factor} = \frac{\pi}{2\sqrt{2}} \approx 1.1107}$$

6) Explain the following terms:-
a) cycle b) Amplitude c) Time period d) frequency
e) Instantaneous value f) Peak to peak value.

a) A cycle is one complete repetition of an AC waveform. It includes one +ve half & one -ve half. After one cycle, the waveform returns to its starting point & shape.

b) Amplitude is the maximum value of the AC quantity measured from the zero level. It indicates how high the waveform goes above the zero line.

Symbol : $V_m$, (or) $I_m$

c) Time period is the time taken to complete one full cycle of the waveform.

Symbol : T
Unit : seconds (s)

relationship (T) = $1/f$

d) frequency is the no. of cycles completed in one second.

Unit : Hertz (Hz)

Relationship : (f) = $1/T$

Eg : $f = 50Hz$

50 cycles per second.

e) Instantaneous value is the value of the AC quantity at a particular instant of time.

Eg :- Sine wave : $v(t) = V_m \sin \omega t$.

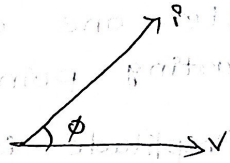
f) peak to peak value is the total height of the waveform from its max +ve peak to max -ve peak.

Peak to peak value = $V_m - (-V_m) = 2V_m$.

7) Explain the power Δle and importance of power factor?

A) Power factor :-

The cosine of angle b/w voltage & current is called power factor.




$$Pf = \cos\phi = \cos(V \wedge i)$$

$$V(t) = V_m \sin\omega t$$

$$i(t) = I_m \sin(\omega t + \phi)$$

Three types of power factors:

1) Active power:-

Power consumed by a resistor if it is supplied by an alternating voltage is called active power (or) real (or) average power.

$$\therefore \boxed{P = I_{rms} V_{rms} \cos\phi}$$

unit : watts.

2) Reactive power :- (Q)

The power developed by the reactor for an alternating voltage is called an reactive power.

$$\therefore \boxed{Q = V_{rms} I_{rms} \sin\phi}$$

unit : VAR. [volt - ampere reactive]

3) Apparent power (s) :-

The product of Rms values of voltage and current is called apparent power.

$$\therefore \boxed{S = V_{rms} I_{rms}}$$

unit : VA (volt - ampere)

* power Δle :-

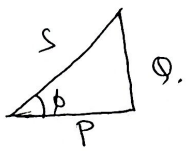
Consider,

$$P^2 + Q^2 = V_{rms}^2 I_{rms}^2 [\cos^2\phi + \sin^2\phi]$$

$$= V_{rms}^2 I_{rms}^2$$

$\therefore \quad S^2 = P^2 + Q^2$

$S = \sqrt{P^2 + Q^2}$.




8) What is Impedance & Explain the impedance $\Delta^{le}$?

A) Impedance (z) :-

Impedance is a complex quantity having real & imaginary values.

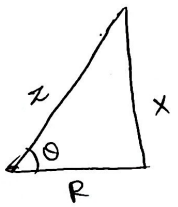
resistance is real value & reactance is imaginary value.

$Z = R + jX$

Z = Impedance

R = resistance

X = Reactance




$Z = \sqrt{R^2 + X^2}$

Impedance $\Delta^{le}$

& $\tan\theta = \frac{X}{R}$

unit = ohm ($\Omega$)

Impedance $\Delta^{le}$ :-

The relation b/w resistance (R), Reactance (X), Impedance (z) :-

